entire modern theory of atoms owes its development to this discovery. A series of other important developments then followed in rapid succession. The two Curies succeeded as early as 1898 in obtaining from uranium a much more strongly radioactive substance, and because of its extremely strong radioactive properties they named it *radium*—that which radiates.

Just about that time, Ernest Rutherford, the real father of modern atomic physics, began to take a hand in its development. He was born in Nelson, New Zealand, in 1871, and died in Cambridge, England, in 1937. Shortly after the announcement of the phenomenon of radioactivity, he found that rays of various types were emitted by radioactive substances, and these rays were distinguished by their different absorbabilities in matter. They are called *alpha*, *beta* and *gamma rays.* The alpha and beta rays are deflected by a magnetic field, which fact indicates that they carry an electric charge—the alpha ray a positive charge, the beta ray a negative one. Gamma rays cannot be deflected—they carry no electric charge.

A more precise study of the alpha rays led to the conclusion that they consisted of particles in rapid movement, each carrying two elementary quanta of positive electricity, and having a mass equal to that of a helium atom (atomic weight 4). The particles which constitute the beta radiation carry only one elementary quantum of negative electricity each, and the mass of each such particle is equal to that of an electron. In other words, beta rays consist of negative electrons in rapid movement. Finally, the gamma rays are identical with x-rays as regards their general properties.

An excellent method was developed by Wilson to make these rays visually detectable. Air containing saturated aqueous vapour is caused to expand suddenly within a chamber, called a *cloud chamber*; it is cooled in consequence and the water vapour therefore becomes supersaturated and has a tendency to condense; though it may remain supersaturated for some time. If now at this moment an alpha particle passes through the chamber, it will tear electrons off the atoms of air along its entire path, leaving positively charged air molecules in its wake. These positively charged air molecules are called *positive ions.* These ions further the condensation of the supersaturated water vapour; in fact, they form ‘condensation nuclei’, around which the water vapour condenses to small droplets. Thus, all along the path of the particle, a very fine track of water droplets is formed, similar to the condensation bands left behind by an aeroplane at high altitudes, thus producing a picture of its path. This phenomenon is shown in Figures 3 (*a*) and (*b*).

The radioactive preparation is under the visible section of the cloud chamber. The tracks of the individual alpha particles can easily be seen and constitute groups of approximately the same length. In Figure 3 (*a*), two groups of different ‘ranges’ are easily discernible.